EFFECTIVENESS OF AN FPQC DESCENT DATUM IS LOCAL ON THE TOTAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's parenthetical '(proof omitted)' attached to the assertion that the constructed data are glueing data as in the glueing section of Schemes. The verification is routine and the surrounding argument is displayed; the parenthetical is reproduced verbatim and the obligation is recorded so that a solver must supply it explicitly.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Open subschemes of X times U that are compatible with descent data are pullbacks of open subschemes of U. Same chapter, Lemma lemma-open-fpqc-covering, invoked at proof lines 9163-9165. Produces the opens U_ij of U_i realising the overlap of the local pieces.
- Morphisms of schemes satisfy fpqc descent, so a compatible morphism descends uniquely. Same chapter, Remark remark-morphisms-of-schemes-satisfy-fpqc-descent, invoked at proof lines 9168-9171. Descends the identity of the overlap to the morphisms phi_ij between the opens U_ij and U_ji.
- Glueing data for schemes and the glueing lemma. Schemes, Section schemes-section-glueing-schemes at proof lines 9172-9174 and Schemes, Lemma schemes-lemma-glue at proof lines 9174-9177. Glues the local schemes U_i along the opens U_ij into a single scheme U over S.

Source reference. Original lemma 03N0: sources/stacks-descent/pinned-original-source.tex, source lines 4318–4318. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 040L: sources/stacks-descent/pinned-original-source.tex, source lines 9118–9118. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01JB: sources/stacks-chow/tags.txt, source lines 2008–2008. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01JA: sources/stacks-chow/tags.txt, source lines 2007–2007. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $X \to S$ be a surjective, quasi-compact, flat morphism of schemes. Let (V, φ) be a descent datum relative to X/S. Suppose that for all $v \in V$ there exists an open subscheme $v \in W \subset V$ such that $\varphi(W \times_S X) \subset X \times_S W$ and such that the descent datum $(W, \varphi|_{W \times_S X})$ is effective. Then (V, φ) is effective.*

Proof. Let $V = \bigcup W_i$ be an open covering with $\varphi(W_i \times_S X) \subset X \times_S W_i$ and such that the descent datum $(W_i, \varphi|_{W_i \times_S X})$ is effective. Let $U_i \to S$ be a scheme and let $\alpha_i : (X \times_S U_i, can) \to (W_i, \varphi|_{W_i \times_S X})$ be an isomorphism of descent data. For each pair of indices (i, j) consider the open $\alpha_i^{-1}(W_i \cap W_j) \subset X \times_S U_i$. Because everything is compatible with descent data and since $\{X \to S\}$ is an fpqc covering, we may apply Lemma 03N0 to find an open $U_{ij} \subset U_i$ such that $\alpha_i^{-1}(W_i \cap W_j) = X \times_S U_{ij}$. Now the identity morphism on $W_i \cap W_j$ is compatible with descent data, hence comes from a unique morphism $\varphi_{ij} : U_{ij} \to U_{ji}$ over S (see Remark 040L). Then $(U_i, U_{ij}, \varphi_{ij})$ is a glueing data as in Schemes, Section 01JA (proof omitted). Thus we may assume there is a scheme U over S such that $U_i \subset U$ is open, $U_{ij} = U_i \cap U_j$ and $\varphi_{ij} = \mathrm{id}_{U_i \cap U_j}$, see Schemes, Lemma 01JB. Pulling back to X we can use the α_i to get the desired isomorphism $\alpha : X \times_S U \to V$. □